FusionSystems

Saturated fusion systems on finite p-groups

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Abstract

The package implements the algorithms of Parker and Semeraro ([PS21]) for saturated fusion systems on finite p-groups in GAP 4: construction of fusion systems from generating automizers and from finite groups, computation of F-conjugacy classes and automizers, a saturation test, the isomorphism test and the classification of all saturated fusion systems on a given p-group. It incorporates the fast search for essential subgroups of Gautam ([Gau26]) and replaces the fusion graph of [PS21] by a lazy, graph free computation.

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Chapter 1

Introduction

1.1 Fusion systems

Let S be a finite p-group. A *fusion system* on S is a category F whose objects are the subgroups of S and whose morphisms $Hom_F(P,Q)$ are injective group homomorphisms containing all those induced by conjugation in S, such that every morphism factors as an isomorphism followed by an inclusion. The fusion system *generated* by subgroups $A_i \leq Aut(E_i)$ ($E_i \leq S$) is the smallest fusion system containing them. By the Alperin–Goldschmidt theorem a saturated fusion system is generated by $Aut_F(S)$ together with the groups $Aut_F(E)$, E running through representatives of the F-conjugacy classes of *essential* subgroups (the fully normalized F-centric subgroups E for which $Out_F(E)$ has a strongly p-embedded subgroup).

In this package a fusion system is therefore given by the data

- S, a p-group given as a pc group,
- automorphisms of S which together with $Inn(S)$ generate $Aut_F(S)$,
- a list of subgroups $E_1,\ldots,E_k$ of S (the essential subgroups) and, for each of them, a list of automorphisms generating $Aut_F(E_i)$.

The fusion system is the one generated by these groups; it is not assumed to be saturated. Whether it is saturated is decided by `IsSaturated` (2.3.1).

1.2 Overview of the algorithms

Essential subgroups. By [Gau26, Theorem 3.6] every essential subgroup of a saturated fusion system on S is S-conjugate to one of the subgroups $C_S(xA/A)$, where A runs through the terms of a central series of S with factors of order p refining the lower central series and contained in $[S,S]$, and xA through the elements of order p of S/A. These candidates are filtered with the tests of [PS21]; all tests are invariant under $Aut(S)$ and the expensive ones (which need $Aut(P)$) are applied once per $Aut(S)$-orbit. The subgroup lattice of S is never computed.

Automorphism groups. Automorphism groups of p-groups are computed with the package autpgrp. If $Aut(P)$ is soluble all computations with its subgroups take place in the pc group returned by `PcGroupAutPGroup`; otherwise a permutation representation is used.

Fusion. Let $B = Aut_F(S) \ltimes S$ and let C be a p'-complement of $Inn(S)$ in $Aut_F(S)$. Every isomorphism in F is a composite of restrictions of elements of $Aut_F(S)$ and of $Aut_F(E_i^s)$, $s \in S$. The

S-conjugacy classes of subgroups form the nodes of a graph which is built lazily: at a node with representative R and for every S-conjugate E^s of an essential subgroup containing R we record the Schreier graph of the orbit of R under $Aut_F(E^s)$, with the arrows conjugated into the class representatives, and the arrows $R \to c(R)$ for the generators c of C. By the Reidemeister–Schreier procedure for groupoids the F-class of R is the connected component of its node and $Aut_F(R)$ is generated by the conjugates of the groups $Aut_S(X)$, X a node of the component, and the loops of the component.

Saturation. By [PS21, Theorem 2.3] (which rests on [BCG+05, Theorem 2.2]) a fusion system generated by $Aut_F(S)$ and the automizers of F-centric subgroups is saturated if and only if $Inn(S)$ is a Sylow p-subgroup of $Aut_F(S)$, every fully normalized F-centric subgroup P is fully automized, and every fully normalized F-centric F-radical P has the *surjectivity property*: for all $P < V \le N_S(P)$ the restriction map $N_{Aut_F(V)}(P) \to N_{Aut_F(P)}(Aut_V(P))$ is surjective. Let T be the set of subgroups of S contained in a B-conjugate of an essential subgroup. T is closed under F-conjugacy and under taking subgroups, and for $P \notin T$ every F-morphism with source P is the restriction of a conjugation by an element of B (the automizers of the essential subgroups never apply). Hence the F-class of P, $Aut_F(P)$ and the extension condition at P are those of the saturated fusion system $F_S(B)$, and the conditions hold. It therefore suffices to run the test over the F-classes of the S-centric subgroups of the essential subgroups. These are found from the subgroup lattices of the groups $E_i/Z(E_i)$, their F-classes and automizers from the lazy graph. This is how the package avoids the fusion graph of [PS21], whose construction (the subgroup lattice of S up to B-conjugacy and the $Aut_F(E)$-orbits of all subgroups of all essential subgroups) is the most expensive part of the original implementation.

Classification. `AllFusionSystems` (3.2.1) follows [PS21, Section 4]: for each p'-subgroup K of $Aut(S)$ up to conjugacy ($Aut_F(S) = Inn(S) \rtimes K$) the proto-essential subgroups are grouped into B-classes; the candidates for $Aut_F(P)$ are the subgroups A of $Aut(P)$ with $Aut_S(P)$ as Sylow p-subgroup and $A/Inn(P)$ having a strongly p-embedded subgroup, one for each $Aut(P)$-class (all conjugates of such a subgroup containing $Aut_S(P)$ as a Sylow p-subgroup are conjugate under $N_{Aut(P)}(Aut_S(P))$), expanded to the conjugates whose Sylow normalizer contains $Aut_{N_B(P)}(P)$, with equality when $N_S(P)$ lies in no B-conjugate of a proto-essential subgroup; the subsets of proto-essentials and the assignments of automizers are run through up to the action of $N_{Aut(S)}(Aut_F(S))$; the resulting systems are tested for $O^p(F) = F$, $O_p(F) = 1$ and saturation and finally reduced up to isomorphism. The automizer candidates are enumerated by the cyclic extension method with the hereditary bound $|H|_p \le |Out_S(P)|$; this is complete whenever GAP can handle the subgroup lattice of $Out(P)$. For elementary abelian P with a huge $Out(P) = GL(n,p)$ the package falls back to the monomial groups of the $Aut_S(P)$-invariant frames (and reports this through `InfoFusionSystems` (3.2.2)), or uses candidates supplied by the user through the option `candidates`.

Chapter 2

Fusion systems

2.1 Construction

2.1.1 FusionSystem

▷ `FusionSystem(`*S*`, `*autSgens*`, `*essentials*`, `*essautos*`)` (function)

returns the fusion system on the pc group *S* generated by $Inn(S)$, the automorphisms in the list *autSgens* (or the group *autSgens* of automorphisms of *S*), and the automorphisms in *essautos*`[i]` of the subgroup *essentials*`[i]` of *S*.

Example

```
gap> S := ExtraspecialGroup( 27, 3 );;
gap> E := First( MaximalSubgroups( S ), M -> IsElementaryAbelian( M ) );;
gap> A := AutomorphismGroup( E );;  n := NiceMonomorphism( A );;
gap> sl := List( GeneratorsOfGroup( DerivedSubgroup( Image( n ) ) ),
>                x -> PreImagesRepresentative( n, x ) );;
gap> F := FusionSystem( S, [], [ E ], [ sl ] );
<fusion system on a group of order 3^3 with 1 essential classes>
gap> IsSaturated( F );
false
```

2.1.2 GroupFusionSystem (for a group and a prime)

▷ `GroupFusionSystem(`*G*`, `*p*`[, `*opts*`])` (function)
▷ `GroupFusionSystem(`*G*`, `*S*`[, `*opts*`])` (function)

returns the fusion system $F_S(G)$ of the finite group *G* on a Sylow p-subgroup *S*. The underlying group of the result is the pc group `Image( IsomorphismPcGroup( S ) )`; the isomorphism is stored in the component `F!.iso` and *G* is the value of the attribute `FusionSystemGroup`. The essential subgroups are searched among the candidates of Gautam's theorem; with `opts := rec( exhaustive := true )` all *S*-centric subgroups are examined instead.

Example

```
gap> F := GroupFusionSystem( PSU( 4, 3 ), 3 );
<fusion system on a group of order 3^6 with 2 essential classes>
gap> OutFSOrder( F );  List( EssentialSubgroups( F ), Size );
4
[ 243, 81 ]
```

```
gap> List( EssentialAutomizers( F ), Size );
[ 3888, 29160 ]
```

2.1.3 UnderlyingGroup (for fusion systems)

▷ UnderlyingGroup(*F*) (attribute)
▷ PrimeOfFusionSystem(*F*) (attribute)
▷ EssentialSubgroups(*F*) (attribute)
▷ EssentialAutomizers(*F*) (attribute)
▷ AutomizerOfS(*F*) (attribute)
▷ OutFSOrder(*F*) (attribute)

the group S, the prime p, the list of essential subgroups, the list of the groups $Aut_F(E_i)$ (as groups of automorphisms), the group $Aut_F(S)$ and the order of $Out_F(S) = Aut_F(S)/Inn(S)$.

2.2 Conjugacy and automizers

2.2.1 AutF

▷ AutF(*F*, *P*) (operation)

returns $Aut_F(P)$ as a group of automorphisms of the subgroup `P` of S (for any subgroup `P`).

2.2.2 FConjugacyClass

▷ FConjugacyClass(*F*, *P*) (operation)
▷ IsFConjugate(*F*, *P*, *Q*) (operation)
▷ FIsomorphism(*F*, *P*, *Q*) (operation)

the list of all F-conjugates of `P`; whether `P` and `Q` are F-conjugate; an isomorphism $P \to Q$ in F (or `fail`).

Example

```
gap> F := GroupFusionSystem( PSL( 3, 4 ), 2 );;
gap> S := UnderlyingGroup( F );;
gap> cls := Filtered( List( ConjugacyClassesSubgroups( S ), Representative ), P -> Size( P ) = 2
gap> Length( cls );  Length( FConjugacyClass( F, cls[1] ) );
9
63
gap> Size( AutF( F, UnderlyingGroup( F ) ) ) / Index( S, Center( S ) );
3
```

2.2.3 IsFCentric

▷ IsFCentric(*F*, *P*) (operation)
▷ IsFullyNormalized(*F*, *P*) (operation)
▷ IsFullyCentralized(*F*, *P*) (operation)
▷ IsFullyAutomized(*F*, *P*) (operation)
▷ IsFRadical(*F*, *P*) (operation)

▷ IsReceptive(F, P) (operation)
▷ HasSurjectivityProperty(F, P) (operation)

the usual properties of a subgroup P of S in F: $C_S(Q) \leq Q$ for all F-conjugates Q of P; $|N_S(P)|$ (resp. $|C_S(P)|$) is maximal in the F-class; $Aut_S(P)$ is a Sylow p-subgroup of $Aut_F(P)$; $O_p(Aut_F(P)) = Inn(P)$; P is fully automized and has the surjectivity property; for all $P < V \leq N_S(P)$ the automorphisms of P extending to F-automorphisms of V are exactly the elements of $Aut_F(P)$ normalizing $Aut_V(P)$.

2.3 Saturation, $O_p(F)$ and the focal subgroup

2.3.1 IsSaturated

▷ IsSaturated(F) (property)
▷ FS_IsSaturatedWithReason(F) (function)

whether the fusion system F is saturated; the second function returns [true] or [false, reason]. The test is the one described in Chapter 1. The essential subgroups of F are required to be F-centric and fully normalized (for the fusion system to be saturated with these essential subgroups); if this fails the result is false.

Example

```
gap> F := GroupFusionSystem( PSL( 3, 3 ), 3 );;
gap> FS_IsSaturatedWithReason( F );
[ true ]
gap> S := UnderlyingGroup( F );;
gap> F1 := FusionSystem( S, [], [ EssentialSubgroups( F )[1] ], [ EssentialAutomizers( F )[1] ] );
gap> FS_IsSaturatedWithReason( F1 );
[ false, "the surjectivity property fails at a subgroup of order 9" ]
```

2.3.2 OpF

▷ OpF(F) (attribute)
▷ IsOpFTrivial(F) (property)
▷ FocalSubgroup(F) (attribute)
▷ IsPPerfectFusionSystem(F) (property)

OpF(F) is the largest subgroup of S normal in S, contained in all essential subgroups and invariant under $Aut_F(S)$ and all $Aut_F(E_i)$; for saturated F this is $O_p(F)$. FocalSubgroup(F) is $\langle [S, Aut_F(S)], [E_i, Aut_F(E_i)] \rangle$, which is the focal subgroup of a saturated fusion system; $O^p(F) = F$ iff it is all of S.

2.4 Isomorphism

2.4.1 IsIsomorphicFusionSystems

▷ IsIsomorphicFusionSystems(F1, F2) (operation)
▷ IsomorphismFusionSystems(F1, F2) (operation)

whether there is an isomorphism $\alpha : S_1 \to S_2$ carrying *F1* onto *F2*, i.e. carrying the generating data of *F1* into *F2* and vice versa; the second function returns such an isomorphism or `fail`. If *F1* and *F2* live on the same group S, α is searched in $Aut(S)$; otherwise an isomorphism of the groups is computed first.

Example

```
gap> F1 := GroupFusionSystem( PSL( 3, 3 ), 3 );;
gap> F2 := GroupFusionSystem( MathieuGroup( 12 ), 3 );;
gap> IsIsomorphicFusionSystems( F1, F2 );
false
```

Chapter 3

Classification

3.1 Essential subgroups and their automizers

3.1.1 RefinedLowerCentralSeries

▷ `RefinedLowerCentralSeries(`*S*`)` (function)
▷ `CentralSeriesCentralizers(`*S*`)` (function)

a central series of the p-group *S* with all factors of order p refining the lower central series, and the list of the subgroups $C_S(xA/A)$ of Gautam's theorem.

3.1.2 ProtoEssentialSubgroups

▷ `ProtoEssentialSubgroups(`*S[, opts]*`)` (function)

returns a record with components `autreps` (representatives of the $Aut(S)$-classes of proto-essential subgroups), `sreps` (representatives of their S-classes), `label` and `path` (`sreps[i]` is the image of `autreps[label[i]]` under the automorphism `path[i][1]` conjugated by `path[i][2]`). With `opts := rec( exhaustive := true )` all S-centric subgroups are examined instead of Gautam's candidates.

Example

```
gap> S := SylowSubgroup( PSU( 4, 3 ), 3 );;  S := Image( IsomorphismPcGroup( S ) );;
gap> r := ProtoEssentialSubgroups( S );;
gap> List( r.autreps, Size );
[ 243, 81 ]
```

3.1.3 ProtoEssentialAutomizerCandidates

▷ `ProtoEssentialAutomizerCandidates(`*S, P[, opts]*`)` (function)

the subgroups A of $Aut(P)$ (as subgroups of the model group of `FS_AutHandle( P )`) containing $Inn(P)$ with $Aut_S(P)$ as Sylow p-subgroup and with $A/Inn(P)$ having a strongly p-embedded subgroup, one for each $Aut(P)$-class. `opts.candidates` may be a function `(S, P)` returning lists of automorphisms (or model subgroups) to be used instead.

3.1.4 BorelSubgroups

▷ `BorelSubgroups(`*`S[, orders]`*`)` (function)

the p'-subgroups K of $Aut(S)$ up to conjugacy, as records with components `model` and `gens` (the generators as automorphisms of *`S`*); $Inn(S) \rtimes K$ runs through the possibilities for $Aut_F(S)$.

3.1.5 HasStronglyPEmbeddedSubgroup

▷ `HasStronglyPEmbeddedSubgroup(`*`G, p`*`)` (function)

whether the finite group *`G`* has a strongly p-embedded subgroup: with T a Sylow p-subgroup this is the case iff $\langle N_G(T), N_G(Q) : Q \leq T, |Q| = p \rangle \neq G$.

3.2 All saturated fusion systems on a p-group

3.2.1 AllFusionSystems

▷ `AllFusionSystems(`*`S[, opts]`*`)` (function)

returns a list of representatives of the isomorphism classes of saturated fusion systems on the pc group *`S`*, by default those with $O_p(F) = 1$ and $O^p(F) = F$. The optional record *`opts`* may have the components `OpTriv` (default `true`), `pPerfect` (default `true`), `OutFSOrders` (a list of admissible orders of $Out_F(S)$), `exhaustive` (default `false`) and `candidates` (a function supplying automizer candidates, see `ProtoEssentialAutomizerCandidates` (3.1.3)). Progress is reported at levels 1 and 2 of `InfoFusionSystems` (3.2.2).

Example

```
gap> L := AllFusionSystems( ExtraspecialGroup( 27, 3 ) );;
gap> Length( L );
4
gap> SortedList( List( L, OutFSOrder ) );
[ 4, 8, 8, 16 ]
gap> ForAny( L, F -> IsIsomorphicFusionSystems( F, GroupFusionSystem( PSL( 3, 3 ), 3 ) ) );
true
gap> Length( AllFusionSystems( ExtraspecialGroup( 343, 7 ) ) );
13
```

The last example recovers the classification of Ruiz and Viruel for 7_+^{1+2} (ten systems realized by finite groups and three exotic ones).

3.2.2 InfoFusionSystems

▷ `InfoFusionSystems` (info class)

the info class of the package.

References

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